PDB-wide identification of biological assemblies from conserved quaternary structure geometry

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Protein structures are key to understanding biomolecular mechanisms and diseases, yet their interpretation is hampered by limited knowledge of their biologically relevant quaternary structure (QS). A critical challenge in inferring QS information from crystallographic data is distinguishing biological interfaces from fortuitous crystal-packing contacts. Here, we tackled this problem by developing strategies for aligning and comparing QS states across both homologs and data repositories. QS conservation across homologs proved remarkably strong at predicting biological relevance and is implemented in two methods, QSalign and anti-QSalign, for annotating homo-oligomers and monomers, respectively. QS conservation across repositories is implemented in QSbio (http://www.QSbio.org), which approaches the accuracy of manual curation and allowed us to predict >100,000 QS states across the Protein Data Bank. Based on this high-quality data set, we analyzed pairs of structurally conserved interfaces, and this analysis revealed a striking plasticity whereby evolutionary distant interfaces maintain similar interaction geometries through widely divergent chemical properties.

The majority of proteins adopt a quaternary structure (QS) by interacting with copies of themselves, thereby forming homomers[1–6]. These protein complexes are present in virtually all biological processes[7–9]. For example, the well-known oncogene p53 is a homomer, as are many transcription factors. In apoptosis, caspase-9 acts as a trigger after homodimer formation[10], and it is the homotrimeric scaffold of the tumor necrosis factor that induces cell death[11]. Metabolic pathways are no exception; for example, most glycolytic enzymes form homomers. Besides being ubiquitous, homomers are common in cellular machines; ATP synthases, proteasomes, chaperones, photosystems, exosomes and histones have all evolved from ancestral homomers[12].

Knowledge of biological QS states is thus key for understanding protein function and evolution. Currently, the richest source of information on protein QS is the Protein Data Bank (PDB), the repository for structural information obtained by X-ray crystallography, NMR spectroscopy, and cryo-electron microscopy[13,14]. Structures solved by X-ray crystallography represent, by far, the majority of the data; more than 120,000 structures have now been deposited. However, X-ray crystallography provides the coordinates of only the asymmetric unit (ASU) within the unit cell of a crystal. At the molecular level, a crystal is formed as an infinite lattice of unit cells (**Fig. 1a**), and a protein complex may be made from one or more ASUs or from parts of one or several ASUs[15]. A critical challenge underlying analysis and interpretation of protein structures is discriminating between fortuitous protein–protein contacts that arise from crystal packing and evolved contacts that make up the biologically relevant QS.

Much computational work has been devoted to addressing this challenge. Several properties of interfaces were found to be important and can be separated into two broad classes. The first class comprises physicochemical and shape properties of protein interfaces and includes size[15–21], planarity[22], atomic packing[19,21,23], predicted binding free energy[15,20], secondary structure[24] and the composition[18,19,21–23,25] or entropy of amino acids[26]. The second class comprises evolutionary conservation, either of individual residues[27–31] or of the geometry of the interaction interface, which can be assessed across different crystal forms[32,33] or across homologs[34,35].

Historically, the protein QS (PQS) server[15] was the first tool to provide information on likely biological assemblies of structures deposited in the PDB and has since been succeeded by another method called PISA[20]. While the predictions of these servers are generally accurate, the hard nature of the problem means that ~15% of QS states from the PDB do not reflect a biologically relevant state[20,36]. This limitation prompted the development of PiQSi[36], which contains 1,434 high-confidence and nonredundant manual annotations. However, manual curation is tedious and cannot be applied at a PDB-wide scale, which motivated us to develop novel automated methods approaching the accuracy of manual curation.

Approaches to superpose structurally related protein chains within arbitrary multi-component complexes have been previously reported[37–39]. Here, we focus on the identification and analysis of homologous QSs sharing the same number of subunits, on the premise that conservation of QS provides a strong indicator for biological relevance (**Fig. 1b**). We developed a heuristic algorithm named QSalign, which structurally aligns complete QSs and infers those that are conserved as being biologically relevant.

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QSalign confirmed the biological relevance of 31,257 homo-oligomeric QS states from the PDB, with an estimated error rate as low as 2.1%. To annotate monomeric proteins, we inverted the QSalign approach and used the absence of homomeric homologs as predictive information of a protein being monomeric itself. We used this method, anti-QSalign, to annotate 46,877 monomeric proteins. In addition to comparing QS states across species, we integrated QS conservation across methods (PISA[20], EPPIC[30], and QSalign/anti-QSalign). The resulting predictor, named QSbio (http://www.QSbio.org), approached the accuracy of manual curation. We finally illustrate the use of QSalign in an analysis of interface properties of structurally similar but evolutionary distant complexes, which reflects the plasticity of interface properties among remote structural homologs.

RESULTS

Inferring biological relevance by multichain superposition of protein quaternary structure

Evolutionary conservation is a powerful means to assess function and can be applied in the context of QS. Previous work indeed reported that structural conservation of interface geometry is a strong indicator of functional relevance[33]. Among dimers, interface conservation directly reflects QS conservation because only a single interface is involved. In higher order oligomers, however, structural similarity cannot be readily inferred from the similarity of pairwise interfaces (**Supplementary Fig. 1**), and this prompted us to assess the structural similarity of full complexes without decomposing them into pairwise interfaces.

The task of superposing full protein complexes presented two main obstacles. A first hurdle arose from the size of QSs, because structural superposition algorithms have been classically applied to single chain proteins of a few hundred residues at most. In contrast, a QS can involve multiple chains that total several thousand residues (**Fig. 2a**). As a solution, we employed the Kpax protein structure alignment algorithm[40], which we found to be fast and robust, even with very large structures. A second obstacle arose from the fact that QS information involves multiple chains, whose coordinates typically appear in PDB files in an arbitrary order. This prompted the development of a heuristic to map chain–chain correspondences (see Online Methods and **Supplementary Fig. 2**). We used this approach to perform pairwise structural superpositions between all homo-oligomeric assemblies from the PDB, and we also included assemblies predicted by PISA. We focused on potential matches by comparing homomers with identical numbers of subunits and domain folds, which resulted in >25 million superpositions. We then developed a strategy to integrate these data and infer the biological QS of protein structures. The overall algorithm is described in detail in the Online Methods as well as in **Supplementary Figures 3** and **4**. Henceforth, we refer to QSalign as the global method that computes structural superposition of protein complexes and uses that information to infer the biological relevance of assemblies in the PDB.

Benchmarking QSalign

We employed a multichain version of the TM-score[41] to assess whether two QS states were similar. To find the optimal threshold and to examine how sensitive results were with respect to the threshold, we benchmarked QSalign using different TM-score cutoff values, from 0.4 to 0.9. For each value, we estimated the accuracy of QSalign annotations by comparing them to manual annotations from PiQSi. As expected, lower cutoff values gave more annotations but also higher error rates (**Fig. 2b**). Above a TM-score of 0.65, however, the error rate remained stable, but coverage decreased. We thus chose a cutoff equal to 0.65, at which 31,257 biological assemblies from the PDB were validated with an error rate of 2.1% (**Fig. 2b**).

Subsequently, we used each validated assembly to search for nonannotated entries with a similar sequence but a different QS. By transitivity, we annotated 11,119 such entries to be potentially incorrect, as illustrated in **Figure 1b**. Benchmarking these annotations against PiQSi showed an error rate estimated at 11.5% (**Fig. 2c**). While this error rate is higher than that of 2.1% obtained when validating assemblies, these annotations are highly valuable because they are specific to erroneous structures.

We next compared the performance of QSalign to two state-of-the-art methods, PISA[20] and EPPIC[30]. Using PiQSi as a benchmark data set, the prediction of dimers using PISA and EPPIC gave results comparable to published values with 13% and 18% error rates, respectively, while we observed an error rate of 4% with QSalign. Among higher order oligomers, QSalign showed the same error rate of 4%, but values for PISA and EPPIC increased to 16% and 32%, respectively. This increase might be due to the fact that high-order oligomers involve several interfaces, and it becomes less likely to predict all of them correctly. We also benchmarked QSalign using a compendium of gold-standard

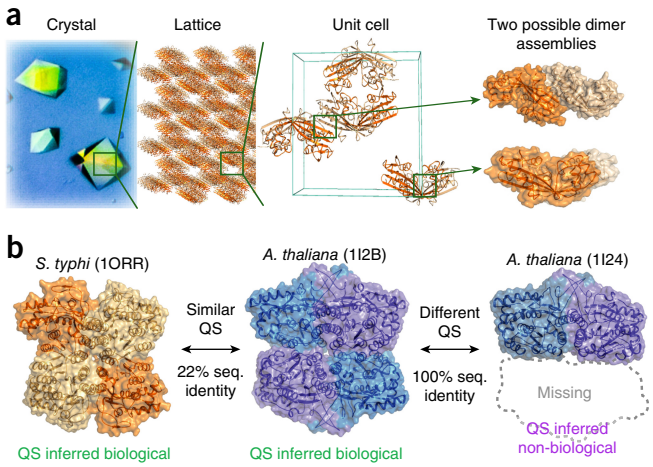

Figure 1 | Quaternary structure conservation across species points to biologically relevant crystal contacts. (**a**) Protein X-ray diffraction requires formation of a crystal. At the molecular level, a crystal is formed by a lattice within which the repeated unit is the unit cell. Here, the unit cell contains eight copies of the protein (PDB code 1EX2) in contact with one another. Identifying biologically relevant contacts among these is challenging. For example, the authors of the structure shown[50] assigned the top dimer as biologically significant, while the method PISA predicts the dimeric form underneath to be so. Searching for homologous structures reveals the latter to be conserved across species (e.g., PDB code 4LU1). (**b**) Tyvelose epimerase is a tetrameric enzyme in *Salmonella typhi* (PDB code 1ORR). A similar tetramer is found in *Arabidopsis thaliana* (PDB code 1I2B, r.m.s. deviation = 3.55 Å), although the sequences of these two tetramers share only 22% identity. Such conservation suggests that both tetramers are biologically relevant. This information enables subsequent correction of entries showing identical sequence but different QS (e.g., PDB code 1I24). Seq., sequence.

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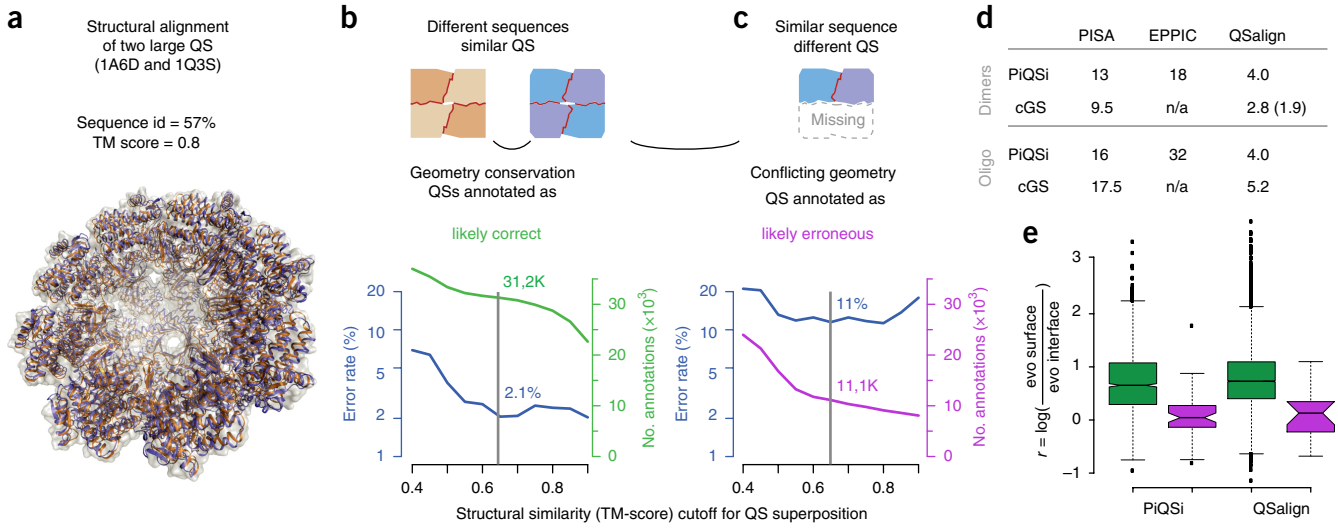

Figure 2 | Quaternary structure superposition and benchmark of predictions. (**a**) Superposition of two large QSs by QSalign (only half of the structure is shown to facilitate visualization). (**b**) We compared QSalign predictions against a manually curated data set from PiQSi[36]. QSalign infers two QSs as correct when their structural similarity exceeds a TM-score cutoff. We scanned different cutoff values and recorded the error rate (blue line) and the total number of QSs annotated (green line) for each. (**c**) QSalign then searches for proteins with conflicting QSs and infers those as erroneous. (**d**) Benchmark of QSalign, PISA and EPPIC using the same data sets. The number in parentheses corresponds to the error rate after discarding a structure for which recent work confirms QSalign prediction. Detailed prediction information for the three methods is given in **Supplementary Data 1.** (**e**) Distribution of the ratio (r) of surface over interface evolutionary rate for residues of QSs predicted to be biological (green) or nonbiological (magenta) in QSalign and PiQSi. Similar distributions of relative interface conservation are observed ($\text{Mean}_{\text{PiQSi}} = 0.70$, $P = 2.5 \times 10^{-15}$; $\text{Mean}_{\text{QSalign}} = 0.77$, $P = 2.6 \times 10^{-6}$ two-sided Wilcoxon test). Boxplots show the interquartile range; whiskers extend to a distance up to 1.5 times the interquartile range or to extreme points if they are closer. evo, evolutionary rate.

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data sets, henceforth the 'cGS' data set (see Online Methods and **Supplementary Table 1**), and we observed similar results (**Fig. 2d** and **Supplementary Data 1**).

QSalign accuracy is supported by residue-level conservation information

In addition to benchmarking QSalign using a manually curated set of protein structures, we carried out a global quality assessment by comparing the evolutionary conservation of amino acids in biological versus nonbiological interfaces. It is known that amino acids present at interaction interfaces tend to be more conserved than surface residues of the same protein[28–31]. Thus, we compared the log ratio, r, of surface over interface evolutionary rate (see Online Methods). If residues are equally conserved in both structural regions, r will be close to 0; whereas higher (or lower) conservation of interface residues will yield positive (or negative) r values. In nonbiological QS states from PiQSi and QSalign, the distribution of r was centered on, or close to 0, which indicated that interface residues behaved like surface residues, as expected. Among biological complexes, r was above 0 on average ($\text{Mean}_{\text{PiQSi}} = 0.70$, $P = 2.5 \times 10^{-15}$; $\text{Mean}_{\text{QSalign}} = 0.77$, $P = 2.6 \times 10^{-6}$), which indicated that residues at biological interfaces are, on average, twice as conserved as surface residues. The similarity of the distribution of r observed among curated structures from PiQSi and among structures annotated with QSalign further reflects the high quality of the annotations derived from the method (**Fig. 2e**). Interestingly, despite the fact that QSalign relies on interface geometry conservation to infer biological relevance, 5.6% of complexes annotated as biologically relevant show r values below 0. Among these, the geometry of the interaction between subunits is conserved, but the residues mediating the interaction are variable.

Annotating monomers with anti-QSalign

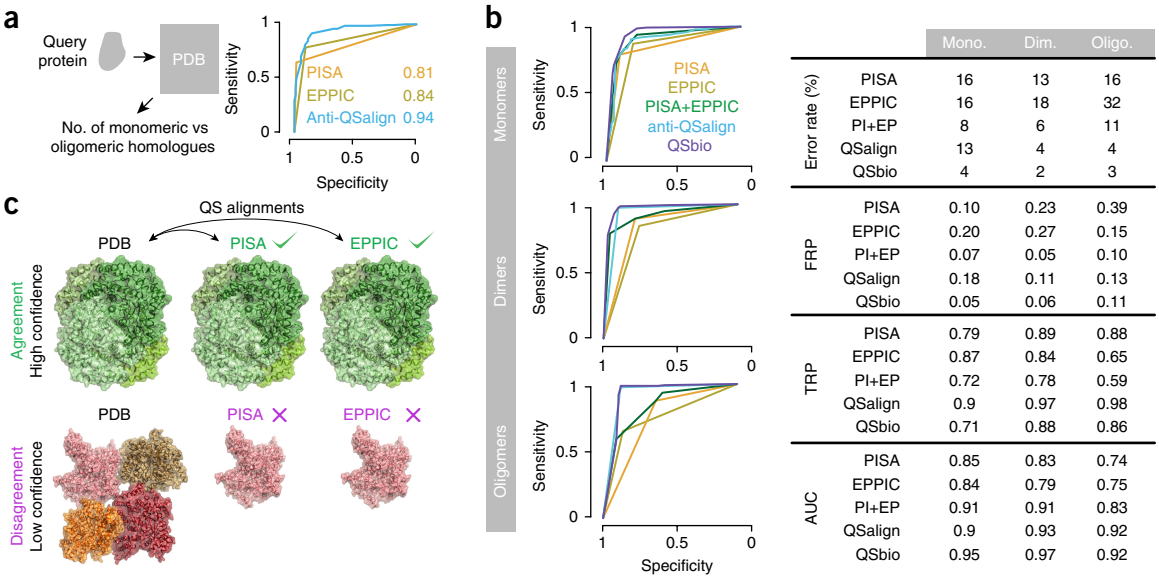
QSalign was unable to annotate monomeric proteins, which by definition do not contain conserved interfaces. Nonetheless, by 'inverting' the QSalign principle, we may consider the absence of interface in homologs to be predictive of monomeric proteins. We call this approach 'anti-QSalign'.

Our predictor was based on counts of nonredundant homologs having or not having a QS. With this strategy, it is expected that protein families with fast-evolving QS will lead to erroneous annotations, while families with conserved QS will allow robust predictions. As for QSalign, the high degree of redundancy in the PDB enabled annotation of a majority of protein structures. We found that ~80% of monomeric proteins (46,877) had at least one homolog in the 30% to 90% sequence identity range and could therefore be classified as being either monomeric or oligomeric by anti-QSalign (see Online Methods).

We benchmarked the predictions of anti-QSalign using both the cGS data set (**Fig. 3a**) and PiQSi (**Fig. 3b**). In both benchmarks, anti-QSalign proved to be a reliable predictor with areas under the curve (AUC) of receiver–operator characteristic (ROC) plot equal to 0.94 (cGS) and 0.9 (PiQSi). These results compared favorably to PISA and EPPIC predictions, which showed maximum AUC values of 0.85.

QSbio integrates predictions and estimates quaternary structure confidence on a PDB-wide scale

Together, QSalign and anti-QSalign enabled annotation of the QS state of over 80,000 structures (**Supplementary Table 2**). Next, we asked whether QS conservation across methods could be used to create a metapredictor with even higher accuracy and coverage. We employed the structural superposition heuristic of QSalign to compare QS states across PISA and the PDB, and we also mapped interfaces annotated by EPPIC onto the PDB assemblies



		Mono.	Dim.	Oligo.
Error rate (%)	PISA	16	13	16
	EPPIC	16	18	32
	PI+EP	8	6	11
	QSalign	13	4	4
	QSbio	4	2	3
FRP	PISA	0.10	0.23	0.39
	EPPIC	0.20	0.27	0.15
	PI+EP	0.07	0.05	0.10
	QSalign	0.18	0.11	0.13
	QSbio	0.05	0.06	0.11
TRP	PISA	0.79	0.89	0.88
	EPPIC	0.87	0.84	0.65
	PI+EP	0.72	0.78	0.59
	QSalign	0.9	0.97	0.98
	QSbio	0.71	0.88	0.86
AUC	PISA	0.85	0.83	0.74
	EPPIC	0.84	0.79	0.75
	PI+EP	0.91	0.91	0.83
	QSalign	0.9	0.93	0.92
	QSbio	0.95	0.97	0.92

Figure 3 | Principle of anti-QSalign and benchmark of QSbio. (**a**) anti-QSalign uses the oligomeric state of homologs to infer whether a protein is monomeric or oligomeric. ROC analysis based the cGS data set (144 monomers and 187 oligomers). The AUC for anti-QSalign, PISA and EPPIC is shown in graph on right. (**b**) Benchmark of individual methods and of their integration into QSbio. ROC curves are shown for each method and separately for monomers, dimers and higher order oligomers. Values of statistics derived from the benchmark are shown in the table. FPR, false positive rate; TPR, true positive rate; AUC, area under the curve. (**c**) Example of two QS states annotated as correct or incorrect in QSbio. The predictions across methods agree for the top QS (green, PDB code 2I56) and disagree for the QS underneath (PDB code 1JQN). Detailed prediction information for the benchmarks of panels **a** and **b** is provided in **Supplementary Data 1**.

(see Online Methods, **Fig. 3c** and **Supplementary Fig. 5**). The comparison of PISA and EPPIC predictions with PDB assemblies enabled consensus predictions to be derived from their agreement or disagreement. We expected that a QS supported by both methods would be more likely to be biologically relevant than a QS supported by only one or neither method.

Among the structures annotated by QSalign or anti-QSalign, QSbio relied on all three methods. For other structures, QSbio combined only PISA and EPPIC. We assessed the performance of QSbio and saw that integrating predictions gave substantial improvements over each method taken individually. Integrating PISA and EPPIC (without QSalign/anti-QSalign) improved AUC values by 0.06, 0.08 and 0.08 for monomers, dimers and oligomers to reach values of 0.91, 0.91 and 0.83, respectively (**Fig. 3b**). A notable improvement is seen among dimers, where the false positive rate is above 0.20 for either PISA or EPPIC, but it decreases to 0.05 when both methods agree.

For structures also annotated by QSalign or anti-QSalign, the performance of QSbio was remarkable; among monomer and dimers, the AUC reached 0.95 and 0.97, respectively (**Fig. 3b**). Among higher order oligomers, the predictive power of QSalign was already maximal (AUC = 0.92) and only marginally improved in QSbio.

Overall, QSbio provides QS annotations with a quality approaching that of manual curation for 80% of monomeric and 70% of homo-oligomeric structures, and it provides high-quality annotations for the rest of the structures. We estimated error rates for each class of prediction within the benchmark, and from these error rates we derived five levels of annotation confidence: very high, high, medium, low and very low, corresponding to estimated error rates of 0–2%, 2–5%, 5–15%, 15–50% and 50–100%. The numbers of structures assigned to each category were, respectively: 51,050; 18,217; 14,995; 14,335 and 11,499 (**Supplementary Data 2**). These annotations will allow the scientific community to select high-confidence QSs to carry out structural analyses. Equally important, the low-confidence classes pinpoint structures in the PDB that may require correction.

Plasticity of interfaces

Interaction interfaces place constraints on the structure, chemistry and evolution of proteins. Structurally, subunits may be required to maintain a precise orientation relative to each other[42]. Chemically, a specific composition at the interface is needed for binding[18,19,21–23], and mutations generally decrease the interaction affinity[43]. As a result, on average, interfaces are hydrophobic, they are enriched in specific sets of amino acids, and their residues are more conserved than those found at the surface[27–31].

There is, however, a large degree of variability around these average trends; some interfaces can be as hydrophilic as protein surfaces, while others can be less conserved than protein surfaces. It can be hypothesized that such variability reflects different structural requirements found across different protein families. In that case, structurally similar interfaces should exhibit similar physicochemical properties. Alternatively, specific protein families may not require specific physicochemical properties of their interface. In that case, physicochemical properties of interfaces would be free to diverge despite the need to maintain the subunit interaction geometry. To resolve this, we compared the variability of interface properties within versus across protein families.

As an initial case study, we analyzed six dimeric alcohol dehydrogenases, which show high structural similarity, with a mean TM-score among pairs equal to 0.81. In these enzymes, NAD (or NADP) is used as a cofactor and is in contact with both subunits, so we expected physicochemical and evolutionary properties of the dimeric interfaces to be conserved. We analyzed three interface properties (interface propensity, interface hydrophobicity and the ratio of interface to surface conservation) among the six

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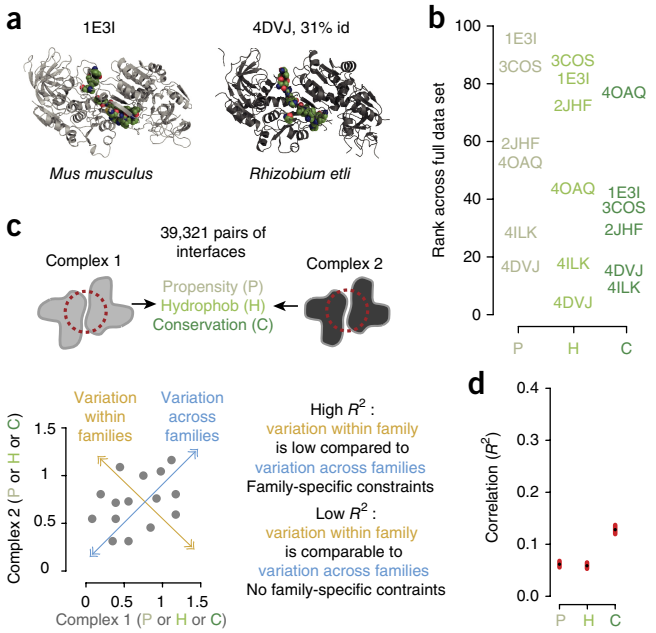

Figure 4 | Protein interfaces are plastic. (**a**) We compared the interfaces of structurally similar protein complexes and exemplify interface plasticity in dimeric alcohol dehydrogenases. Two homologous structures sharing 31% sequence identity are shown. The interface, shown with spheres, is conserved structurally but not chemically. (**b**) Rank of interface properties for six alcohol dehydrogenases denoted by their PDB code, where 0 and 100 correspond, respectively, to the lowest and highest values seen in any complex from the data set. (**c**) We evaluated interface plasticity across the entire data set. The correlation coefficient (R^2) of a given property for pairs of structures reflects whether that property is constrained in a family-specific manner. If pairs of homologous interfaces exhibit similar values for the properties examined, a high R^2 value is expected. In contrast, if there are no family-specific constraints, homologous interfaces will be as different from each other as two random interfaces can be, and R^2 will be low. (**d**) Correlation coefficients calculated among 21,434 pairs of structurally similar interfaces with sequences sharing less than 30% sequence identity. Dot plots from which the correlation values were calculated are shown in **Supplementary Figure 6**. Error bars show the 95% confidence interval.

members of the family. We then compared how these properties varied relative to the entire data set of 39,321 pairs (**Fig. 4a**). We observed that interface properties across the six enzymes spanned nearly the entire range seen in the data set (**Fig. 4b**). For example, the human enzyme (PDB code 3COS) has a highly hydrophobic interface, while that of *Rhizobium etli* (PDB code 4DVJ) is predominantly hydrophilic. Similarly, in terms of conservation, the interface of the enzyme from *Escherichia coli* (PDB code 4ILK) is 3.9 times more conserved than the rest of the surface, while its yeast homolog (PDB code 4OAQ) shows interface conservation similar to that of the rest of the surface. Thus, these dimers do not need their interface to meet strict family-specific physicochemical requirements. Rather, their interfaces exhibit nearly as much variability as is sampled among dimers in general.

Is such interface plasticity specific to this alcohol dehydrogenase family, or is it general to all protein families? To answer this question, we derived a data set of 39,321 pairs of homologous interfaces based on the results of QSalign. We then assessed the within-family variability of each property relative to the variability across families. If variability within families is small compared with the variability seen across families, the correlation coefficient will be close to one. However, if the variability within families is as large as the variability across families (as we observed for alcohol dehydrogenases), the correlation coefficient will be close to zero (**Fig. 4c**). When comparing interface properties among homologs sharing 80% to 90% sequence identity, we observe high correlations, with R^2 values equal to 0.83, 0.55 and 0.54 for interface conservation, hydrophobicity and propensity, respectively (**Supplementary Fig. 6**). However, the fact that close homologs share interface properties is expected, because these homologs have not had sufficient evolutionary time to diverge. As sequence identity decreases among the homologs being compared, so does the similarity between their interfaces, to the point where the interfaces within two homologous assemblies are nearly as different as two random interfaces. Indeed, for the three interface properties we considered, the correlation among distant homologs gets close to zero (**Fig. 4d**), even though they share the same interface geometry and overall structure. Thus, this result generalizes previous observations made on heteromeric and transient interactions[44,45] and shows that homo-oligomeric interfaces, which are often large and stable, are nevertheless highly plastic.

DISCUSSION

We introduced two complementary approaches, QSalign and anti-QSalign, which allow QS prediction based on conservation across evolution. While we focused on annotating monomers and homo-oligomers, QS conservation could, in principle, also be applied to hetero-oligomers. Such an extension will require additional developments, because complexes with different numbers of subunits will need to be compared, and because subunits of different sizes make a global measure of structural similarity hard to interpret.

The main limitation of QSalign and anti-QSalign is their requirement for homologous structures, but the high species coverage of the PDB allowed annotation of ~80% and ~70% of monomeric and homo-oligomeric structures, respectively (**Supplementary Table 2**). We anticipate that these fractions will increase with time as more structures are solved. Moreover, increasing the number of high-quality QS annotations will be important for detailed structural analyses such as the examination of structural properties of disease-causing mutations[46,47] or the investigation of determinants of protein evolution[48,49].

In addition to using QS conservation across evolution, we employed QS comparison to integrate predictions of different methods with QSbio, which considerably improved the prediction accuracy for monomers and dimers. A noteworthy feature of QSbio is the confidence estimates that will facilitate the interpretation of QS information both for individual structures and for PDB-wide data sets serving in structural bioinformatics analyses.

METHODS

Methods, including statements of data availability and any associated accession codes and references, are available in the online version of the paper.

Note: Any Supplementary Information and Source Data files are available in the online version of the paper.

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AUTHOR CONTRIBUTIONS

S.D. and E.D.L. designed and performed the experiments. D.W.R. adapted the Kpax algorithm to enable the calculations. S.D. and E.D.L. wrote the manuscript with input from D.W.R. All authors corrected and approved the final manuscript.

COMPETING FINANCIAL INTERESTS

The authors declare no competing financial interests.

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ONLINE METHODS

Data sets. The data set of protein structures on which all analyses were performed is based on 3DComplex as of March 2015 (ref. 2) and consists of 110,498 'biological assemblies' defined by the Protein Data Bank. Note that the total number of structures annotated in QSbio is slightly lower (110,096) because EPPIC lacks some annotations. The data set is available on the 3DComplex (version 5) website: http://shmoo.weizmann.ac.il/elevy/3dcomplexV5/Home.cgi.

For each structure, we also included the top prediction from PISA[20] as of April 2015. EPPIC assignments of interfaces were kindly provided by G. Capitani (Paul Scherrer Institute, Switzerland) in October 2015.

Structural superposition of quaternary structures. Structural superposition was carried out using Kpax (version 3)[40]. Briefly, Kpax first aligns two structures by placing all possible pairs of 5-mer fragments at the coordinate origin by exploiting the tetrahedral geometry of their central Cα atoms and by using dynamic programming (DP) to find the optimal combination of pairs of fragments and hence pairs of central Cα atoms. The aligned Cα atoms are then superposed by least-squares fitting, and the resulting superposition is refined using further rounds of DP on the atomic Gaussian overlap of the superposed Cα pairs. Each Gaussian represents the volume of one Cα atom. Thus, if two Cα atoms overlap perfectly, they contribute one volume unit to the overall superposition score; and if two aligned protein chains superpose perfectly, then the total Gaussian overlap score directly corresponds to the volume of their overlaid Cα atoms. Kpax can align structures with multiple chains, but only in the given chain order. To find the maximum aligned overlap volume between two multichain oligomers, it is necessary to determine the best chain order of one structure with respect to the other. We therefore employed a two-step heuristic procedure to avoid the computational cost of running Kpax repeatedly on different chain order permutations. In this procedure, a first alignment and superposition is carried out. The coordinates of the superposed QSs are then analyzed to map corresponding chains across the two structures. Scores between chain pairs are calculated and correspond to the number of Cα atoms of one chain closer than 2 Å from any Cα from another chain. The highest ranking chain pairs provide correspondences that are used to rewrite the PDB file in matching chain order. The reordered coordinates are then used in a second round of structural superposition, which yields the final TM-score. Pseudocode describing this procedure is provided in a **Supplementary Note**. An executable version of the structural superposition program is available at https://github.com/elevywis/QSalign.

Protein Data Bank superposition of quaternary structures. The annotation process of QSalign first required structural superposition of homo-oligomers. To save computation time, we carried out structural alignments between potential matches only—that is, between pairs of structures sharing structural homology, as reflected in their SCOP or PFAM domain architecture, or sharing sequence homology (>30% sequence identity). Ultimately, we measured the structural similarity of 25,965,020 QS pairs and recorded the TM-score for each, which was stored in a MySQL database table. We provide the pseudocode for the comparison of one query structure with multiple target structures as a function "QSalign" in a **Supplementary Note**.

Annotation procedure. The execution of structural alignments was followed by an inference process described in the pseudocode "QSinfer" (**Supplementary Note**). Briefly, a query QS was annotated as correct if another structurally similar QS (TM-score > threshold) was found for a homolog with less than 80% sequence identity. Pairs of QSs sharing more than 80% sequence identity were not considered in order to reduce the risk of the same crystal packing being formed on account of protein-surface similarity. The annotation process was carried out for each symmetry group separately, starting with those containing larger numbers of subunits. This condition ensured that a lower order oligomer (e.g., a dimer) would be annotated as correct only if no evidence for high-order oligomerization (e.g., a tetramer) was found. Once all QSs from a symmetry group were processed, those annotated as correct were used to search for possible errors. The overall strategy is described in the pseudocode "QSpropagate" (**Supplementary Note**). The correction step consists of identifying proteins with an identical sequence and a structure different from that of a QS annotated as being correct (**Supplementary Figs. 3** and **4**). Then, the annotation depends on the number of subunits of the QS to be annotated (QS2) and the QS that is supposedly correct (QS1). We distinguish three cases: (i) if QS2 has more subunits than QS1, we analyze the consistency of QS2 given its number of subunits and its symmetry. We call QS2 consistent when the number of subunits is expected from the point-group symmetry (e.g., dimer for C2, tetramer for D2, etc.). Consistent structures are flagged as 'ambiguous' and are not annotated as errors, while inconsistent structures are annotated as errors. (ii) If QS2 has the same number of subunits as QS1, we infer that QS2 is incorrect. (iii) When QS2 has fewer subunits than QS1, we infer that interfaces are missing from QS2, which is also annotated as incorrect.

To optimize the TM-score threshold at which two QSs were considered equivalent, we carried out several full cycles of annotation (e.g., inference + propagation steps) using different TM-scores cutoff (**Fig. 2**) and benchmarked the results after each cycle based on PiQSi[36].

Benchmarking the annotations of QSalign for TM-score optimization. After each cycle of the annotation procedure, we counted the number of structures in the following categories: TP, annotated as correct by QSalign and PiQSi; FP, annotated as correct by QSalign and as incorrect by PiQSi; FN, annotated as incorrect by QSalign and as correct by PiQSi; and TN, annotated as incorrect by both QSalign and PiQSi. We calculated the error rate of 'correct' annotations by the false discovery rate FDR = FP/(TP + FP) and the error rate of 'incorrect' annotations by the false omission rate FOR = FN/(FN + TN). The data set of structures on which these rates were calculated was filtered at a level of 90% sequence identity. In addition, we used only high-confidence annotations from PiQSi that did not contain the tag "probable." The resulting number of structures from PiQSi available for the benchmark was 1,434.

Annotating monomers with anti-QSalign. We annotated monomers based on the QS state—either monomeric or oligomeric—of

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known homologs. The underlying assumption of this approach is that QS is generally conserved during evolution[51], so structures homologous to a true monomer should be more likely to be monomeric than oligomeric. Thus, for a particular protein sequence, the enrichment of monomeric over oligomeric homologs was used to derive a probability score to be monomeric. These probabilities were estimated based on PDB structures not found in any benchmark, as follows. We defined six bins of 'numbers of homologs': 0, 1, 2, 3–4, 4–8, and >8; and each protein was assigned to two such bins, one for the number of monomeric homologs and the other for the number of oligomeric homologs. We then recorded frequencies of monomers in each bin combination; for example, 4.4% of monomers have three or four monomeric homologs and a single oligomeric homolog. Then, we recorded the same frequencies for oligomers; for example, 0.08% of oligomers have three or four monomeric homologs and a single oligomeric homolog. The frequencies were subsequently converted into probabilities; for example, we estimated that a protein has a probability equal to 4.4/(4.4 + 0.08) = 0.98 to be monomeric if it has three or four monomeric homologs and a single oligomeric homolog. This process is illustrated in **Supplementary Figure 7**. We considered proteins sharing a minimum of 30% and a maximum of 90% sequence identity to be homologs. We also imposed a minimum of 60% overlap of the sequence alignment relative to the longest of the two proteins being aligned.

Integrating quaternary structure information with QSbio. QSbio compares QSs from the PDB with predictions of EPPIC, PISA, and QSalign/anti-QSalign when available. The comparison with PISA was carried out by structural superposition, as described above, and pairs of QSs with a TM-score above 0.9 were considered identical. To compare PDB and EPPIC QSs, we mapped interface groups from EPPIC onto PDB structures. For the mapping, we decomposed each PDB assembly into pairs of contacting chains, and each pair was superposed onto the interfaces from EPPIC to identify matching pairs (**Supplementary Fig. 5**). We then derived a weighted score for a PDB structure based on its annotation by the different methods. For monomers, the score was calculated as: $S_{MONO} = 0.4 \times \text{antiQSalign} + 0.4 \times \text{PISA} + 0.2 \times \text{EPPIC}$. For oligomers (including dimers), the score was calculated as: $S_{OLIGO} = 0.7 \times \text{QSalign} + 0.2 \times \text{PISA} + 0.1 \times \text{EPPIC}$. In the benchmark (**Fig. 3b**), we considered positives as those with $S_{MONO} \geq 0.8$ or $S_{OLIGO} > 0.8$ and the negatives as those having a score equal to 0. When the annotation of (anti)-QSalign was not available, the score was renormalized to 1.0, but the same 2:1 weights for PISA and EPPIC were used. We estimated the expected error rates (err) in QSbio by: $\text{err} = \text{FPR} \times \text{fN} / (\text{FPR} \times \text{fN} + \text{TPR} \times \text{fP})$, where TPR and FPR are the true and false positive rates measured in each score class, and fP and fN are the (unknown) fractions of correct and incorrect structures in the PDB. We aimed for conservative error rates, so we used a relatively high value for fN, with fN = 0.2 and fP = 0.8. Importantly, the error rates estimated for each class also depend on the benchmark data set and on the assumed pattern of errors in the PDB. In that respect, their absolute value should be interpreted with care. We do expect, however, that the accuracy rank of the different classes will be robust. To reflect this limitation, as well as to simplify the use of these predictions by end users, we created five confidence categories: 'very high', 'high', 'medium', 'low' and 'very low', corresponding respectively to estimated error rates of 0–2%, 2–5%, 5–15%, 15–50%, and 50–100%.

Comparative benchmarks against PiQSi. We compared the set of annotations obtained with QSalign and anti-QSalign to predictions derived from PISA and EPPIC. PiQSi annotations refer to specific PDB structures. Therefore, we could directly compare the annotations of PiQSi to the annotations of PISA and EPPIC for the corresponding structure. Positives and negatives were counted when the methods supported (positive, P) or did not support (negative, N) a particular PDB structure. These predictions, together with PiQSi annotations, allowed a contingency table to be created and performance statistics to be derived. ROC curves were plotted using R[52] and the pROC package[53]. The numbers of structures (broken down by oligomeric state) used in this comparative benchmark are given in **Supplementary Table 1**, and details of predictions are provided in **Supplementary Data 1**.

The conditions to count positives and negatives depend on the oligomeric state of the PDB structure considered. First, for monomeric proteins, the conditions for counting positives and negatives were:

anti-QSalign: the 'monomer probability score' associated with the PDB code was above (positive) or below (negative) 0.4.

PISA: no assembly was predicted (positive), or an assembly was predicted (negative).

EPPIC: no biological interface was predicted (positive), or at least one was (negative).

Combined PISA + EPPIC: both methods gave a positive prediction (positive) or both gave a negative prediction (negative). Disagreements (19% of the cases) were not used to calculate the error rate.

QSbio: $S_{MONO} >= 0.8$ (positive), or $S_{MONO} = 0$ (negative). Intermediate scores (27% of the cases) were not used to calculate the error rate.

Second, for oligomers with two or more subunits, the conditions for counting positives and negatives were:

QSalign: the structure is annotated as correct (positive) or incorrect (negative).

PISA: the structure matches (positive, TM-score > 0.9) or does not match the PDB structure (negative, TM-score < 0.9).

EPPIC: when biological interfaces are mapped onto the PDB structure, they either maintain all subunits in contact (positive), or they do not (negative).

Combined PISA + EPPIC: both methods gave a positive prediction (positive), or both gave a negative prediction (negative). Disagreements (20% and 35% of the cases for dimers and oligomers, respectively) were not used to calculate the error rate.

QSbio: $S_{OLIGO} > 0.8$ (positive), or $S_{OLIGO} = 0$ (negative). Intermediate scores (16% and 16% of the cases for dimers and oligomers, respectively) were not used to calculate the error rate.

Comparative benchmarks against the compendium gold-standard data set cGS. We assembled a second gold standard based on previously published data sets. We used data sets published by Bahadur *et al.*[54], Ponstingl *et al.*[18] and Duarte *et al.*[55], which gave a total of 338 structures including 144 monomers, 137 dimers, and 57 oligomers (**Supplementary Table 1**, **Supplementary Data 1**). We call the so-obtained data set the

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compendium gold standard (cGS). Note that the data set from Duarte *et al.*[55] was used for 'monomers/nonmonomer' information only.

An important difference to note between cGS and PiQSi is that we only used 'PDB code identifiers' for the cGS and not for actual structures. Therefore, we used the 'number of subunits' to assess predictions. For monomers, the conditions for counting positives and negatives were identical to those used in the PiQSi benchmark. For oligomers, however, they differed as follows:

QSalign: the number of subunits predicted by QSalign is equal to the number of subunits in the benchmark data set (positive) or is different (negative).

PISA: the number of subunits in the PISA assembly is equal to the number of subunits in the benchmark data set (positive) or is different (negative).

EPPIC: we could not infer the number of subunits solely from interface information, and so we were unable to include EPPIC in this benchmark.

Interface properties. We selected pairs of homologous dimers matching the following criteria: the TM-score was above 0.65, structure resolution was below 2.5 Å, structures were representatives of a nonredundant set filtered at 90% sequence identity, and structures were annotated as biological by QSalign. All properties were calculated on interface core residues, as defined in Levy[56]. To calculate sequence conservation, each protein was used to search for homologs in Uniref90 (ref. 57) and those homologs with >40% sequence identity and 80–100% coverage were kept to create a multiple alignment using MUSCLE[58] with default parameters. Alignments with fewer than five sequences were discarded. We subsequently used rate4site with default parameters[59] to obtain amino-acid-specific evolutionary rates. Rates were scaled by subtracting the minimum score to each alignment. Interface core and surface evolutionary rates were obtained by averaging the scores of residues composing each region. The relative conservation of interfaces was obtained by the ratio of evolutionary rate of the interface relative to the surface (interface/surface), and these were used in **Figure 4** and **Supplementary Figure 6.** In **Figure 2e**, we used log(surface/interface) to obtain symmetric distributions. In **Figure 2e**, $r > 0$ indicates that the interface is more conserved relative to the surface. Interface hydrophobicity was calculated by the ratio of hydrophobic residues present at the interface. Interface residue propensity was calculated by averaging individual amino-acid propensities taken from Dey *et al.*[60] and normalizing them by the size of the interface.

Statistics. The evolutionary rate distributions (**Fig. 2e**) were compared using a two-sided Wilcoxon test. The confidence intervals of **Figure 4d** were calculated based on Fisher's Z transform as implemented in the cor.test function of the R statistical language.

Code availability. The code implementing the heuristic procedure to be used with the Kpax algorithm for QS alignment is available at https://github.com/elevywis/QSalign.

Life Sciences Reporting Summary. Further information on experimental design and reagents is available in the **Life Sciences Reporting Summary**.

Data availability statement. All the annotations and confidence categories derived from this work are available for download at http://www.QSbio.org (for the most up-to-date information) and in **Supplementary Data 2**.

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Experimental design

1. Sample size

Describe how sample size was determined.	Sample size was determined by the available structures in the Protein Data Bank and in the 3DComplex database

2. Data exclusions

Describe any data exclusions.	All the available structures were included

3. Replication

Describe whether the experimental findings were reliably reproduced.	NA

4. Randomization

Describe how samples/organisms/participants were allocated into experimental groups.	NA

5. Blinding

Describe whether the investigators were blinded to group allocation during data collection and/or analysis.	NA

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For all figures and tables that use statistical methods, confirm that the following items are present in relevant figure legends (or in the Methods section if additional space is needed).

n/a	Confirmed	
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☐	☒	The test results (e.g. *P* values) given as exact values whenever possible and with confidence intervals noted
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Software

Policy information about availability of computer code

7. Software

Describe the software used to analyze the data in this study.	Data analyses were carried out with custom Perl and R scripts. The core engine of the work for structural superposition of quaternary structures is available on Github. The predictions resulting from the method are available at www.QSbio.org

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